

Retail Business Performance & Profitability Analysis

Introduction

Retail businesses generate large volumes of transactional data that can be analyzed to improve profitability and overall business performance. This project focuses on analyzing retail sales data to identify profitable categories, product performance, regional trends, and sales patterns. The insights obtained from the analysis can help businesses make better strategic decisions and improve operational efficiency.

Abstract

The retail dataset was imported into SQL and cleaned by handling missing and null values. SQL queries were used to calculate profit margins by category and sub-category. Python (Pandas and Seaborn) was used to perform correlation analysis, generate heatmaps, identify fast-moving and slow-moving products, and analyze monthly sales trends. Tableau dashboards were developed to visualize sales, profit, category performance, product profitability, and regional performance.

Tools Used

- SQL
- Python (Pandas, Seaborn)
- Tableau

Steps Involved in Building the Project

1. Imported retail transaction data into SQL.
2. Cleaned missing and null records from the dataset.
3. Calculated profit margins by category and sub-category using SQL queries.
4. Performed correlation analysis using Python.
5. Created heatmaps to understand relationships between business metrics.
6. Identified fast-moving and slow-moving products based on sales performance.
7. Generated monthly sales trend analysis using Python visualizations.

8. Built Tableau dashboards with KPI cards and interactive charts.
9. Created the visualizations
10. Added filters for Region and Product Category to improve dashboard interactivity.
11. Derived business insights and recommendations from the analysis.

Key Insights

1. Sales and profit showed a positive relationship based on correlation analysis.
2. Fast-moving products contributed significantly to overall sales performance.
3. Slow-moving products may require promotional strategies to improve sales and profitability.
4. Certain categories and sub-categories generated higher profits than others.
5. A small number of products contributed a major share of total profitability.
6. Some products consistently generated losses and require review of pricing and sales strategies.
7. Monthly sales analysis revealed fluctuations in sales performance across different periods.
8. Regional analysis highlighted differences in sales and profit contribution across locations.
9. Due to the absence of inventory-related fields, sales trends were used as an alternative measure to evaluate product movement and performance.

Conclusion

This project successfully analyzed retail business performance using SQL, Python, and Tableau. The analysis provided valuable insights into sales trends, profitability, product performance, and regional performance. The Tableau dashboard enabled interactive exploration of business metrics and supported data-driven decision-making.

Although inventory turnover analysis could not be performed directly due to the absence of inventory-related fields in the dataset, sales-based analysis effectively identified fast-moving and slow-moving products. Overall, the project demonstrates how data analytics can be used to improve business performance, profitability, and strategic planning.